

Selected topics

Search and Sort



Search functions on an array (1/3)

Compute the minimum (maximum) value of an array (a vector V or a matrix M) over the given dimensions

```
minimum(array, dims)  
maximum(array, dims)
```

```
julia> minimum(V)  
julia> minimum(M)  
julia> minimum(M,dims=1)  
julia> minimum(M,dims=2)
```

```
julia> maximum(V)  
julia> maximum(M)  
julia> maximum(M,dims=1)  
julia> maximum(M,dims=2)
```

Search functions on an array (2/3)

For an array (a vector V or a matrix M), return the indices of the minimum (maximum) elements over the given dimensions

```
argmin(array, dims)
```

```
argmax(array, dims)
```

```
julia> argmin(V)
```

```
julia> argmin(M)
```

```
julia> argmin(M,dims=1)
```

```
julia> argmin(M,dims=2)
```

```
julia> argmax(V)
```

```
julia> argmax(M)
```

```
julia> argmax(M,dims=1)
```

```
julia> argmax(M,dims=2)
```

Search functions on an array (3/3)

For an array (a vector V or a matrix M), return the index or key of the first element of A for which predicate returns true. Return nothing if there is no such element.

```
findfirst(predicate, array)
```

```
julia> findfirst(isequal(3), M)
```

index i, j of the first element of M for which $M[i, j]=3$

```
julia> findfirst(iseven, M)
```

```
julia> findfirst("Linz", "Hello Linz")
```

See also `findlast`, `findprev`, `findnext`

Sort functions on a vector (1/2)

Returns a sorted copy of a vector V

```
sort(vector)
```

```
julia> sort(V)
```

Returns the vector V sorted

```
sort!(vector)
```

```
julia> sort!(V)
```

Option to reverse the sorting order

```
sort(vector, rev=true)
```

```
julia> sort(V, rev=true)
```

Sort functions on a vector (2/2)

Return a permutation vector I that puts $V[I]$ in sorted order

```
sortperm(vector)  
sortperm(vector, rev=true)
```

```
julia> V=[2,7,4]  
julia> p=sortperm(V)  
julia> V[p]  
julia> p=sortperm(V, rev=true)  
julia> V[p]
```

